

Introduction to AI

Evolution of AI from Simple Concepts

Evolution of AI

- 1950s: Birth of AI
- Machine Learning
- Deep Learning
- Generative AI



Types of AI

NARROW AI (WEAK AI)



Specialized in
one task only

Example: Siri,
Google Translate

GENERAL AI (STRONG AI)



Can think, reason
& learn like humans

Example:
Still under research

SUPER AI

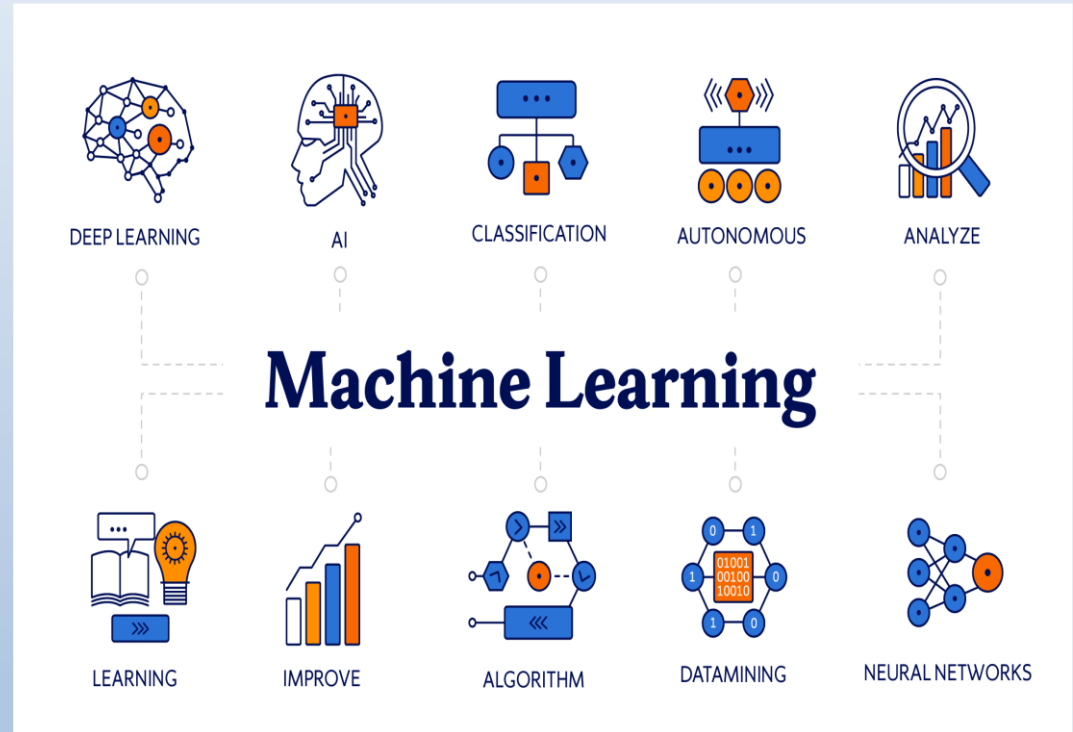


Surpasses human
intelligence

Example:
Theoretical / Future

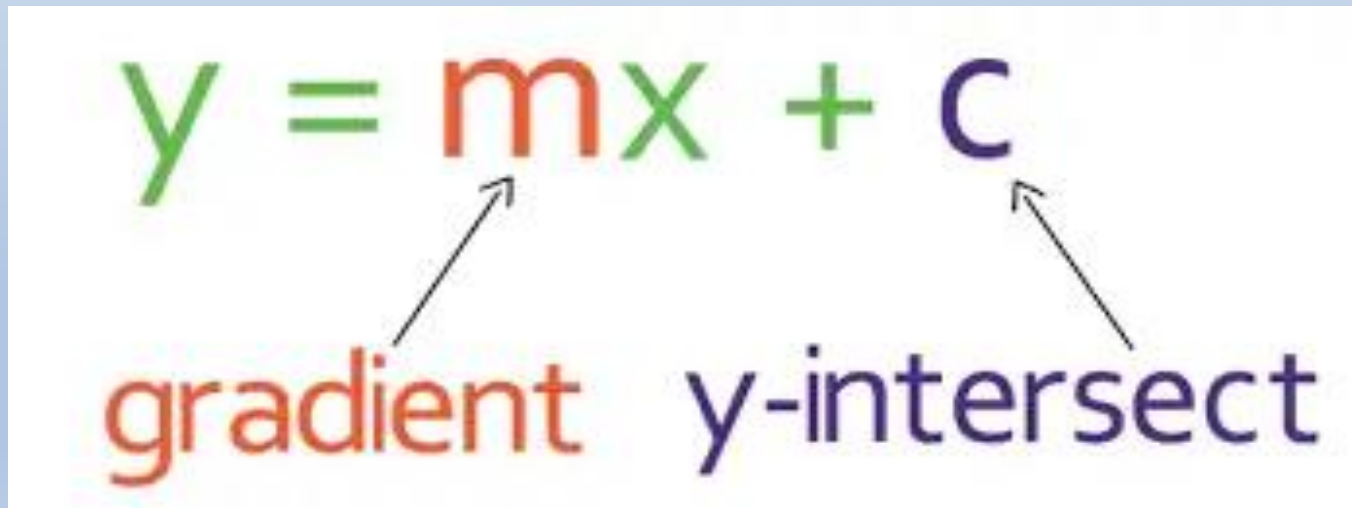
Machine Learning

- Subset of AI
- Learns from data
- Makes predictions



Linear Regression

- Equation: $y = mx + c$
- Predicts continuous values



The diagram shows the linear regression equation $y = mx + c$ in a large font. Below the equation, the word "gradient" is written in red, and the word "y-intersect" is written in dark blue. Two arrows point from the labels to the corresponding parts of the equation: one arrow points from "gradient" to the coefficient m , and another arrow points from "y-intersect" to the constant c .

$$y = mx + c$$

gradient y-intersect

Linear Regression Example

$$X=[1,2,3,4,5] \quad y=[3,5,7,9,11]$$

Numerator:

$$\sum (X - X_{\text{mean}}) \cdot (y - y_{\text{mean}})$$

Denominator:

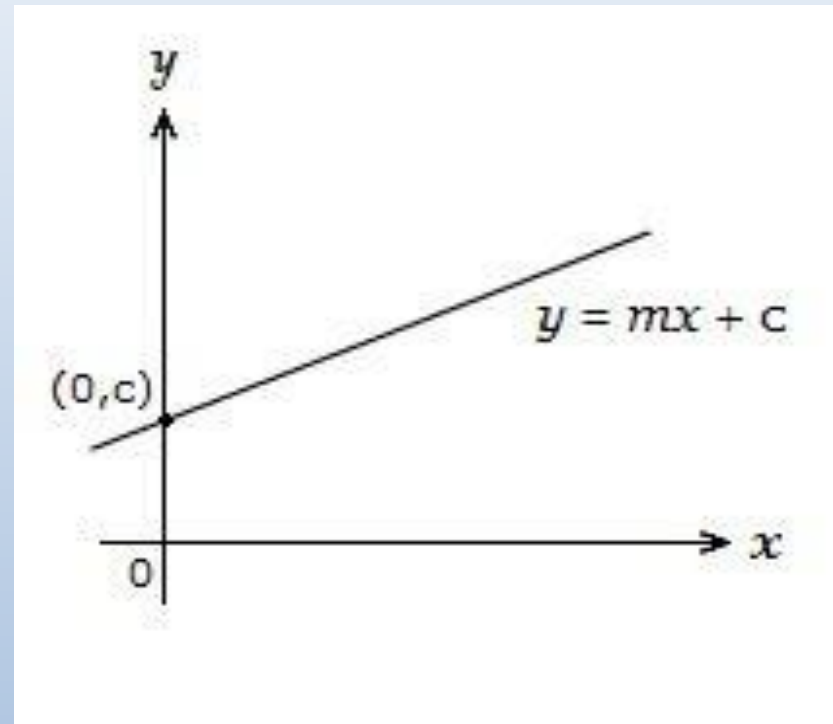
$$\sum (X - X_{\text{mean}})^2$$

slope (m):

$$m = \text{Numerator} / \text{Denominator}$$

intercept (c):

$$c = y_{\text{mean}} - m \cdot X_{\text{mean}}$$



$$\text{Result: } y=2x+1$$

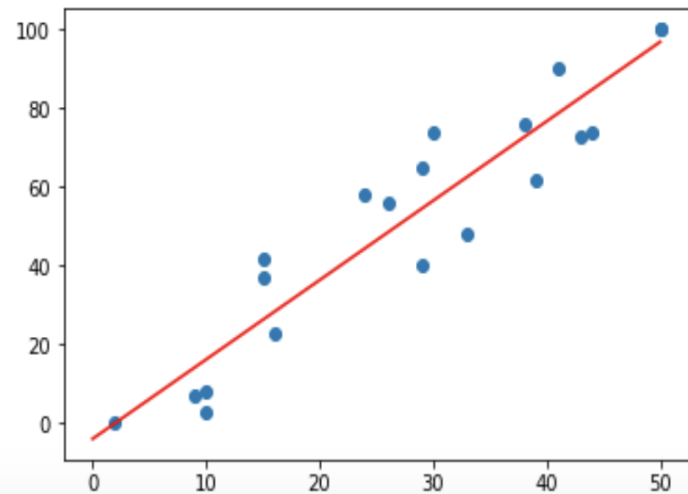
Python Experiment

- Train Linear Regression
- Predict outputs
- Plot regression line

Step #6 - Dataviz

```
In [12]: x_lin_reg = range(0, 51)  
y_lin_reg = predict(x_lin_reg)  
plt.scatter(x, y)  
plt.plot(x_lin_reg, y_lin_reg, c = 'r')
```

Out[12]: [<matplotlib.lines.Line2D at 0x1a19bff5550>]

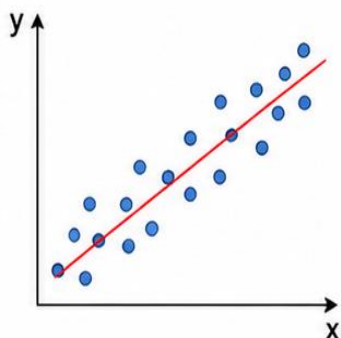


From Regression to ML

STEP 1

LINEAR REGRESSION

Start with a simple linear model to predict outcomes.

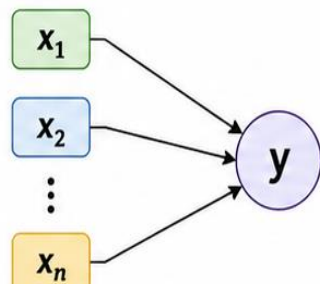


Predicts the relationship between one input (x) and output (y).

STEP 2

MULTIPLE LINEAR REGRESSION

Extend to multiple features (inputs) to improve predictions.



Equation

$$y = m_1x_1 + m_2x_2 + \dots + m_nx_n + c$$

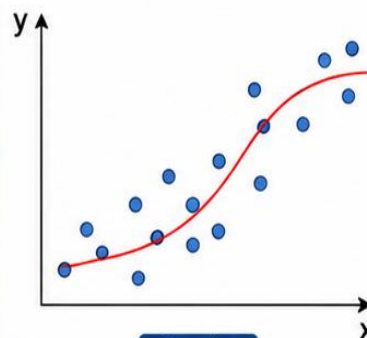


Considers multiple inputs to give better predictions.

STEP 3

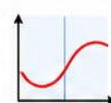
POLYNOMIAL REGRESSION

Introduce non-linearity by adding polynomial terms.



Equation

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$



Captures non-linear patterns in the data.

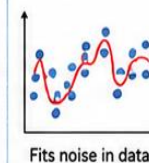
STEP 4

REGULARIZATION

Add techniques like Lasso and Ridge to prevent overfitting.

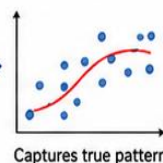
WHY REGULARIZATION?

OVERFITTING



Fits noise in data

GOOD GENERALIZATION



Captures true pattern

RIDGE (L2)



Penalizes large coefficients by squaring them.

LASSO (L1)

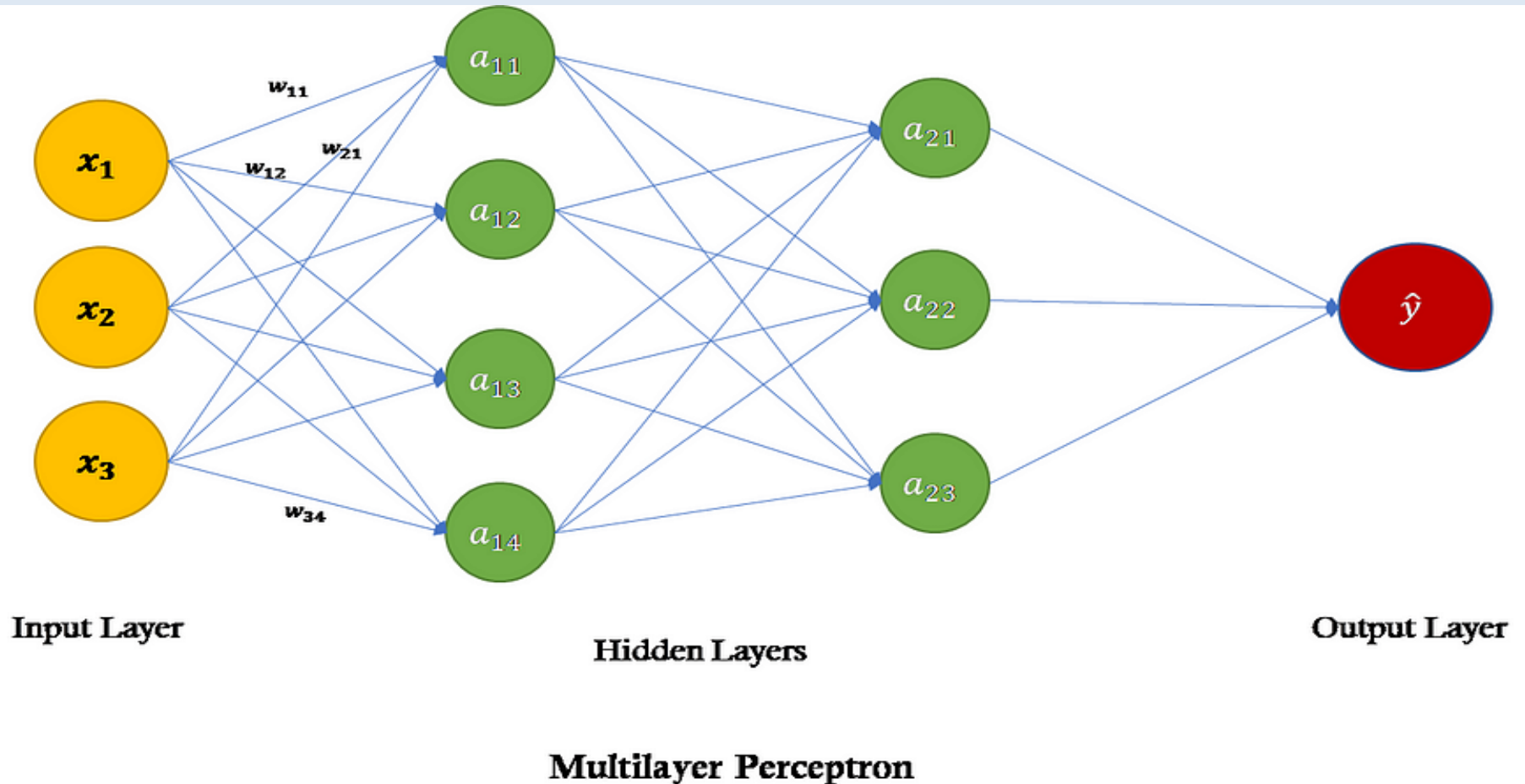


Penalizes large coefficients by absolute value.

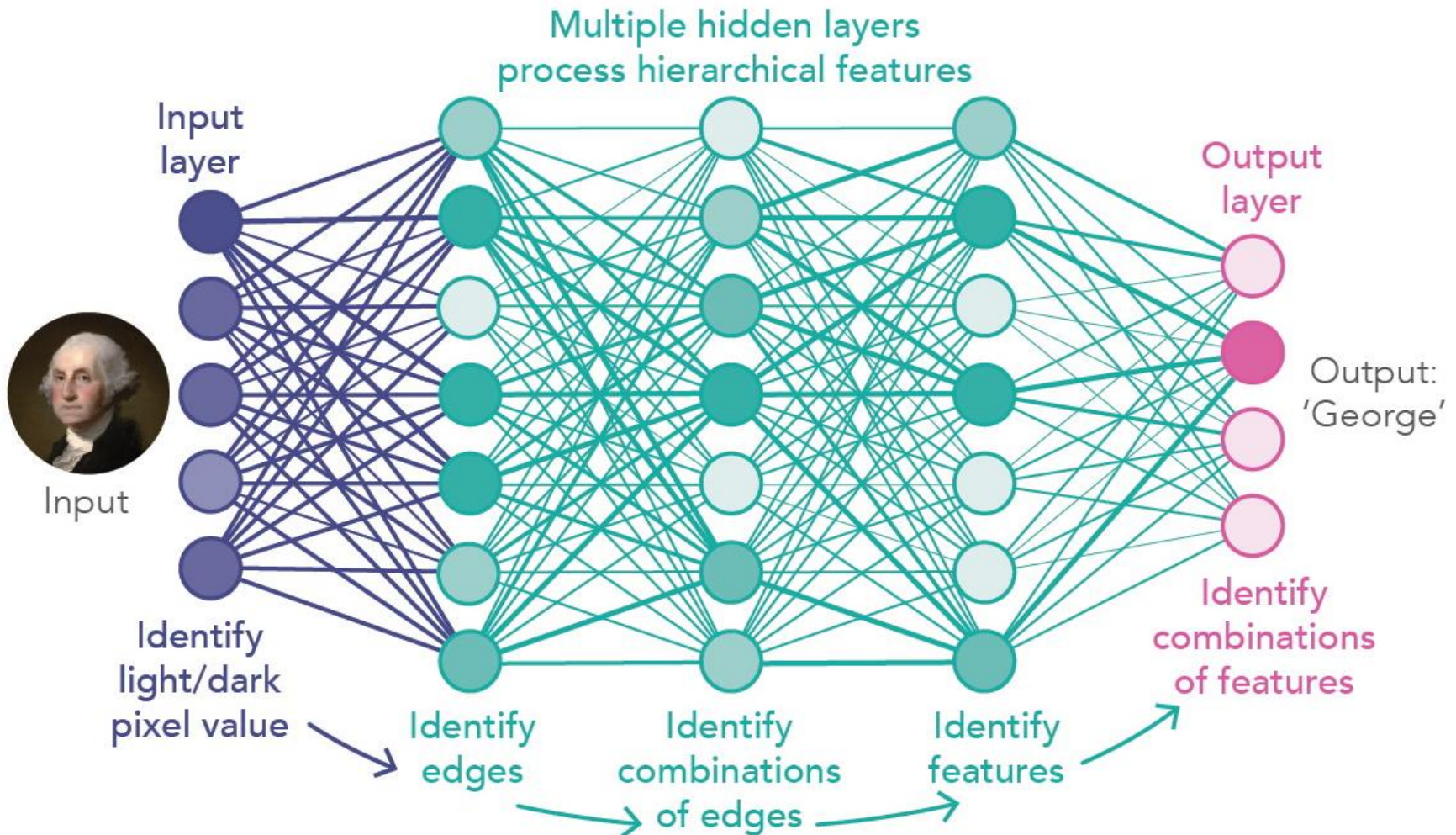


Helps prevent overfitting and improves model performance on unseen data.

Neural Networks

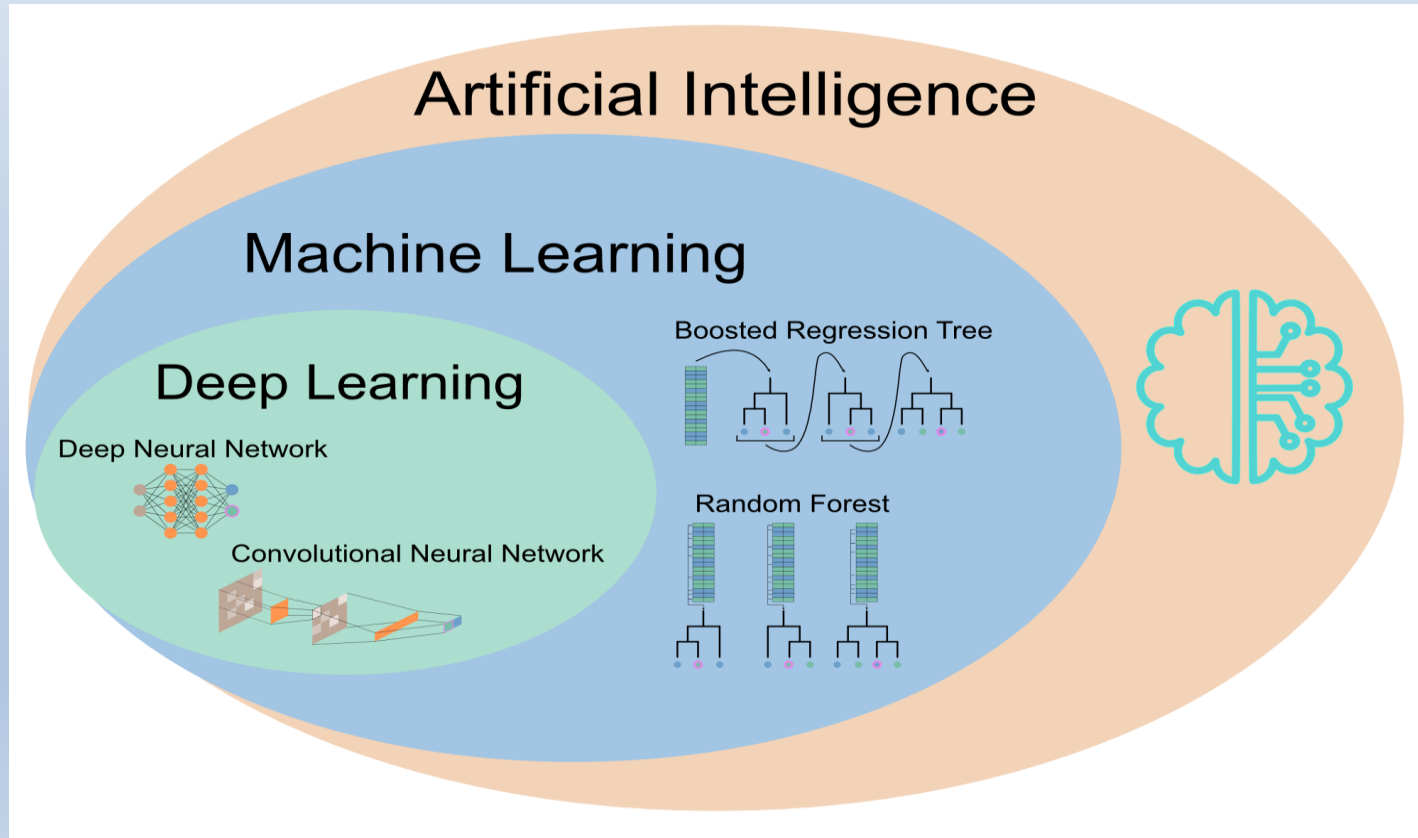


ANN to Deep Learning

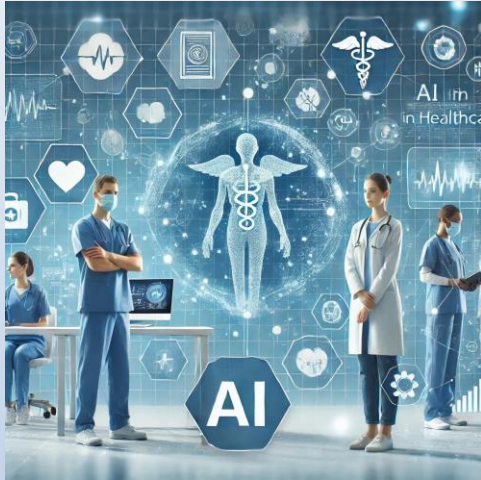


ML vs Deep Learning

- ML: Feature engineering
- DL: Automatic feature learning



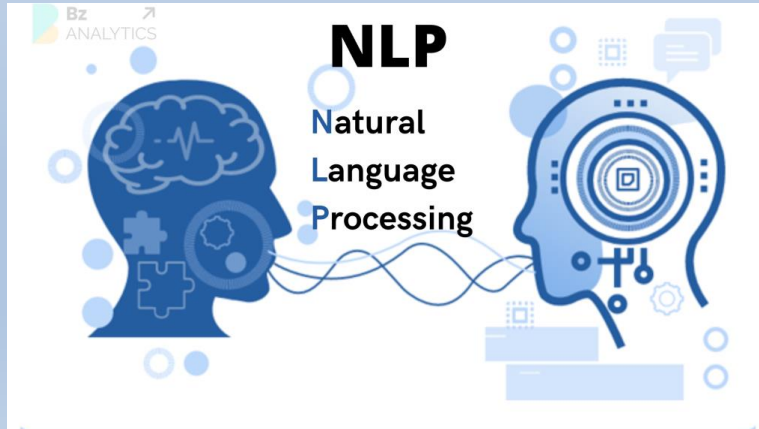
AI Applications



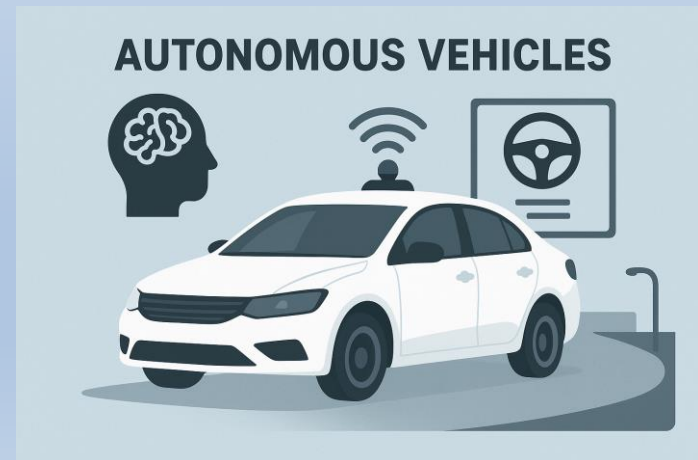
Healthcare



Finance



NLP



Autonomous Vehicles

Conclusion

- Machine Learning helps AI learn from data and make better decisions.
- It is used in many fields like healthcare, finance, education, and transportation.
- Keep exploring Machine Learning to understand how it is shaping the future of AI.